

Ujjwal Prakash

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Professional Summary

High-performing Full-Stack Engineer, Research Co-author, and Technical Instructor with a proven track record of architecting scalable, high-concurrency systems. Promoted to Senior Software Engineer & Team Lead for the official IIT Kanpur placement portal (RAS), managing a team of 4 developers and a 37K+ line production codebase for 20,000+ students. Co-authored peer-reviewed deep learning research for ISRO, accepted for a prestigious oral presentation at IEEE IGARSS 2026. Expert in Go, Node.js, and React/Next.js, delivering high-availability cloud applications with 99.9% uptime.

Education

Indian Institute of Technology (IIT), Kanpur | *B.Tech in Biological Science and Bioengineering* | Expected: May 2028 - **Cumulative Performance Index (CPI):** 6.7 / 10.0 - **Relevant Coursework:** Algorithms & Data Structures, Programming Concepts, Computational Methods - **Certifications:** Data Structures & Algorithms (Udemy)

Aklank Public School, Kota | *CBSE (XII)* | 2024 - **Aggregate Percentage:** 82.0%

Holy Cross School | *CBSE (X)* | 2022 - **Aggregate Percentage:** 94.0%

Research & Publications

Co-author | *IEEE IGARSS 2026, Washington D.C.* | May 2026 - **TRINETRA: Text-Guided Remote-Sensing Interpretation via Ensemble-Based Transformers** - Developed a novel vision-language ensemble framework in collaboration with the Indian Space Research Organisation (ISRO). - Engineered the full-stack system architecture, real-time database sync, and low-latency API integrations to interface with the core Vision-Language Models. - Accepted as a prestigious **Oral Presentation** (Session WE3.R15: Vision-Language Models for Remote Sensing) at the flagship IEEE International Geoscience and Remote Sensing Symposium.

Experience

RAS (IITK Placement Portal) | *Senior Software Engineer & Team Lead* | Kanpur, India | Aug 2024 – Present - **Engineering Leadership:** Promoted to Lead Engineer to oversee full system architecture and maintain a 37,000+ line production codebase (Go/Gin, Next.js, PostgreSQL, Docker) serving **20,000+ students** and top-tier recruiters (Google, Microsoft, Tower Research Capital, Optiver, JPMC). - **Team Management:** Direct and mentor **4 junior developers**, accelerating feature delivery by **25%** and reducing post-deployment bugs by **35%** through standardized code reviews and design sessions. - **Automated QR System:** Spearheaded the design and development of an institutional QR-based attendance tracking app, integrating it with RAS portal APIs to dynamically sync data for **20,000+ students**; mentored 4 engineers during development and coordinated training sessions for 50+ stakeholders on system operations. - **System Stability:** Resolved critical departmental eligibility and resume-tagging microservice bottlenecks, securing **99.9% uptime** and sub-200ms page loads during peak traffic surges. - **Resume Tagging Microservice:** Designed and implemented a resume-tagging service, enabling students to apply to different companies and roles using role-specific resumes. - **System Operations:** Function as primary on-call engineer, implementing secure API gateways, deploying zero-downtime hotfixes for high-stakes placement operations, and optimizing the portal's email notification service to reduce delivery failures. - **Technical Support:** Provided direct technical support to recruiters and students, diagnosing production issues and rolling out hotfixes under real-time constraints.

PhD Placement Portal | *Software Engineer* | IIT Kanpur | 2024 – Present - **API Development:** Architected clean backend APIs and relational database schemas matching PhD candidates with corporate R&D sectors. - **Stakeholder Collaboration:** Translated academic recruitment policies into secure, multi-role access controls and

workflows. - **System Maintenance:** Developed and maintained core features of the production platform, fixing bugs and implementing feature updates in a live environment.

Freelance Developer | *Self-Employed* | 2024 – Present - **Scalable Application Delivery:** Currently engineering a web application designed to scale for **50,000+ users** for an external client. - **End-to-End Solutions:** Delivered multiple full-stack web applications, independently managing requirements elicitation, system architecture, database design, and production deployment. - **Tech Stack & Deployment:** Developed high-performance backends (Go, Node.js) and modern frontends (Next.js) with clean REST APIs; deployed and maintained production systems using Docker. - **Client Collaboration:** Handled direct client communications, project scoping, milestone tracking, and post-delivery system maintenance.

SPO (Students' Placement Office), IIT Kanpur | *Web Executive* | Kanpur, India | 2024 – 2025 - **Core Modules:** Engineered and deployed core Next.js frontend modules, improving Google Lighthouse performance score to **95+** and increasing organic search traffic by **25%**. - **Type Safety:** Architected a type-safe TypeScript migration, reducing runtime production exceptions by **45%** and accelerating development velocity. - **UI/UX & Navigation:** Redesigned the navigation architecture and UI, reducing recruiter drop-off by **30%** and shortening job posting creation times by **20%**. - **Performance Optimization:** Enhanced frontend responsiveness and accessibility, ensuring compatibility across diverse device architectures.

Projects

SPO Critical Incident Investigation | *Security Forensics & Incident Response* - **Incident Management:** Co-led the end-to-end investigation and root cause analysis of a critical system incident, authoring a comprehensive post-mortem report. - **Security Forensics:** Employed advanced forensic techniques and detailed log analysis to successfully track down and identify the specific user responsible for the anomaly. - **Preventive Architecture:** Designed and implemented new security protocols and validation checks to mitigate future risks and ensure long-term system integrity.

ISRO Satellite Image Analysis Platform (TRINETRA) | *Python, FastAPI, Flask, Next.js, TypeScript, Supabase, Modal, Qwen-VL, LLaMA 3.1* - Designed and built an end-to-end satellite image analysis platform using VLMs, YOLO-OBB, and SAM, supporting captioning, grounding, and multi-type VQA. - Won the **Silver Medal** at Inter IIT Tech Meet 14.0 for the ISRO problem statement (out of 100+ teams). - Architected a two-layer backend: Flask orchestration + GPU-hosted FastAPI inference on Modal, enabling scalable and modular ML serving. - Optimised inference pipelines with LoRA fine-tuned Qwen-VL, LLaMA 3.1, and 4-bit quantisation, balancing latency and GPU memory usage. - Developed a Next.js + TypeScript frontend with real-time chat, geospatial comparison, and map-based analysis; integrated Supabase for storage, authentication, and real-time database sync.

Placement Interview Management System (PIMS) | *Go, Gin, Next.js, PostgreSQL, Docker, Nginx* - Architected and deployed a real-time interview operations platform to replace manual, Excel-based workflows, achieving immediate adoption by 1,000 - 2,000 daily active users from day one. - Engineered a Global Availability Engine with real-time conflict detection and seamless RAS portal sync, eliminating data silos and saving the organization 100s of hours in manual operational overhead daily. - Mentored **4 engineers** through the full development lifecycle while coordinating training sessions for **50+ placement coordinators and faculty** on system operations.

Research Lab Portal | *NestJS, Prisma ORM, PostgreSQL, AWS, React, Tailwind CSS, TypeScript* | *Ongoing* - Architecting a centralized portal/ecosystem to automate research lab applications, designing the system to support a **30,000+ campus community** across IIT Kanpur. - Designing relational data models using Prisma ORM to guarantee referential integrity, optimizing database indexing to reduce local query latency to sub-100ms (a **40% speedup**). - Implementing a scalable cloud infrastructure using AWS for deployment and Supabase for real-time synchronization and authentication. - Developing a high-fidelity, responsive frontend using Tailwind CSS and TypeScript, prioritizing professional UI/UX standards to ensure a seamless experience for academic stakeholders.

Cipher — AI-Powered Productivity OS | *Next.js, React, TypeScript, PostgreSQL, Docker, GitHub Actions* - Engineering a personal productivity platform with task management, goal tracking, focus modes, and mood boards — designed around individual workflows. - Integrating an LLM pipeline that accepts natural-language feature descriptions, generates production-ready code, and auto-deploys via a CI/CD pipeline on merge — zero manual intervention from describe to deploy. - Building a REST API with proper auth, environment management, and secrets handling for a fully self-hosted production deployment.

2D Battle Arena — Real-Time Multiplayer Combat Game | *Go, WebSockets, TypeScript, Canvas API, Redis, PostgreSQL* - Implementing a fixed-timestep authoritative game loop in Go that syncs player position, health, and

combat actions across all connected clients in real time. - Building a Canvas-based renderer with client-side prediction and server reconciliation to mask network latency and deliver smooth 60fps gameplay. - Engineering robust WebSocket connection lifecycle handling — graceful drops, automatic reconnection, and state recovery — optimized for sub-50ms round-trip latency.

RAS.DEVCAMP Portal | *Next.js 16, React 19, TypeScript, Tailwind CSS 4, Lenis, Netlify* - Built the course portal for a 10-week full-stack bootcamp at IIT Kanpur SPO, serving **57+ enrolled students** with week-by-week curriculum modules, structured assignment specs with multi-part rubrics, a curated resource library, and a live cohort tracker with deduplication logic. - Designed a premium editorial-style UI with glassmorphic announcement cards, Lenis smooth-scrolling, responsive sidebar navigation, and animated micro-interactions — all hand-built without component libraries. Deployed on Netlify with automated builds.

Technical Skills

- **Programming Languages:** Go, TypeScript, JavaScript, Python, C, C++, R, CSS, HTML
 - **Backend & Cloud:** Node.js, Express, NestJS, Gin (Go), Fastify, FastAPI, Flask, Prisma ORM, AWS, Supabase, Docker, RESTful APIs, WebSockets, Redis, Modal
 - **Frontend & Design:** React, Next.js (Next.js 16, React 19), Tailwind CSS (Tailwind CSS 4), Lenis, Canvas API
 - **Databases:** PostgreSQL, MongoDB
 - **Developer Tools & ML:** Git, GitHub Actions, Nginx, PyTorch/Deep Learning (VLM integration), YOLO-OB, SAM
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Leadership & Achievements

- **RAS.DEVCAMP — Full-Stack Developer Bootcamp, IIT Kanpur SPO** | *Instructor & Curriculum Designer* | Jan 2026 – Present
 - Designed and delivered a 10-week full-stack/backend engineering bootcamp under IIT Kanpur’s Students’ Placement Office, enrolling **57+ students** (100 undergraduate students total across programs).
 - Delivered **12+ lectures** teaching core backend and full-stack concepts from JavaScript/Go fundamentals through Docker, WebSockets, and CI/CD, with live coding sessions and real-time Q&A.
 - Provided hands-on mentorship through structured assignments with multi-part grading rubrics, debugging sessions, and one-on-one support to help students ship capstone projects end-to-end (including Cipher and 2D Battle Arena).
- **ACA (Association for Computing Activities) Summer Project, IIT Kanpur** | *Project Mentor* | Summer 2026
 - Mentored **20 undergraduate students** in building end-to-end full-stack applications (including Cipher and 2D Battle Arena) live on the internet; successfully reviewed and ratified all project completions.
 - Designed a curriculum focused on web foundation basics, Node/Express, Go backend game loops, WebSockets, Git, databases, Docker, and CI/CD pipelines.
- **Astronomy Club, IIT Kanpur** | *Secretary* | 2024 – 2025
 - Managed club operations, directed astronomy observation camps, and mentored an intensive summer research project for **15 students**.
- **Prayas (IITK Social Initiative)** | *Volunteer English Teacher* | 2024 – Present
 - Taught English and fundamental academics to Class 6 students under IIT Kanpur’s student-run educational outreach program.
- **Scholastic Achievements & Competitions:**
 - **Silver Medalist:** Inter IIT Tech Meet 14.0 for the ISRO Satellite Image Analysis problem statement (out of 100+ teams).
 - **State-level Qualifier:** Rank 8 in the International Mathematics Olympiad (IMO).